AWS Cost Reduction Report

KMF Website — Next.js + Strapi Application

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Billing period analysed: June 2026 — Total $134.99 | Status: Analysis only — no code changed

$134.99 JUN 2026 (ACTUAL)	**~$30** MAY 2026	**+350%** MONTH-OVER-MONTH	**$25–45** REALISTIC TARGET/MO

1. Executive Summary

The KMF site is a **Next.js 14** frontend backed by a **Strapi CMS**, deployed with Docker on **AWS EC2** and fronted by an **API Gateway** and **Load Balancer**, with media in **S3** inside a **VPC with a NAT Gateway**. The bill jumped from roughly **$30 in May** to **$134.99 in June** — a ~4.5× increase. The increase is driven by infrastructure that is over-engineered for a mostly-static marketing website that serves large, unoptimised images and videos.

Bottom line: With no loss of functionality, this site can realistically run for **$25–$45 / month** — a saving of roughly **$90–$110 / month (~$1,100–$1,300 / year)**. The single biggest lever is putting a **CloudFront CDN** in front of S3 and **compressing the media**; the second is **removing API Gateway + NAT Gateway**, which add ~$30/mo for little benefit on this workload.

2. Where the money is going (June 2026)

#	SERVICE	COST	PRIORITY	MOST LIKELY CAUSE
1	**S3**	$44.15	HIGH	Large media (157 MB of video, 19 MB images in `public/`) + Strapi uploads served *directly* from S3 with no CDN → high data-transfer-out and GET request charges on every page view.
2	**EC2-Instances**	$36.38	HIGH	Likely an over-sized / always-on instance running Next.js SSR + Strapi. 252 raw `<img>` tags with no optimisation push render and bandwidth load onto the instance.
3	**API Gateway**	$21.07	HIGH	API Gateway charges per request + data transfer ($3.50/M + data). Using it to front a website is an expensive pattern — the ALB already does this job.
4	**Tax**	$20.59	AUTO	Proportional to the bill (~18% GST). Falls automatically as the items above drop. Not separately actionable.
5	**VPC**	$10.01	MEDIUM	Almost certainly a **NAT Gateway** (~$0.045/hr + $0.045/GB). Rarely needed for a single public web instance.
6	**Elastic Load Balancing**	$1.70	LOW	Partial-month ALB. Inexpensive; keep it — but it makes API Gateway redundant.
7	EC2-Other	$1.09	LOW	EBS volumes / snapshots / elastic IP. Minor.

TOTAL	**$134.99**

* The May→June spike strongly suggests a re-architecture in June (S3 + API Gateway + NAT Gateway all appeared) and/or a traffic increase against un-cached, heavy media. May was essentially just one EC2 instance.

3. Evidence found in the project

OBSERVATION	WHY IT COSTS MONEY
`public/video` = **157 MB** (reel-6.mp4 16 MB, marketing1.mp4 13 MB, our-product.mp4 12 MB, …)	Every uncached video play streams MBs of S3/EC2 egress at $0.09/GB. A handful of reels can dominate the S3 bill.
`src/images` = **386 MB** (portfolio 193 MB, product 98 MB, homeImages 59 MB)	Bloated bundle; many of these become large static assets shipped to every visitor.
252 raw `<img>` tags; **0** uses of `next/image`	No automatic resizing, WebP/AVIF conversion or lazy-loading → full-size originals downloaded every time.
`next.config.js` has **two** **`module.exports`** (the first is overwritten) and no `images` config	Misconfiguration; image-optimisation domains/formats are never set, so optimisation is effectively off.
`.next` build dir = **4.1 GB** (4.0 GB is `.next/cache`)	Indicates a heavy build with lots of media churn; inflates EBS/instance disk and image-optimisation cache.
Git repo pack = **1.1 GB**	Large binaries committed to git; slows CI/CD and bloats any storage that mirrors the repo.
Strapi CMS backend (`@strapi/blocks-react-renderer` , `NEXT_PUBLIC_STRAPI_URL`)	The Strapi server is a second always-on process — a candidate to right-size or co-locate with Next.js on one instance.

4. Recommended changes (ranked by saving ÷ effort)

① Put CloudFront (CDN) in front of S3 and the site EASY · BIGGEST WIN

Serve all images, videos and static assets through a CloudFront distribution instead of hitting S3 / API Gateway on every request. CloudFront caches at the edge, so repeat views cost almost nothing and data-transfer is cheaper.

- Cuts S3 GET-request and egress charges dramatically (the $44 line).
- Offloads traffic from EC2 and removes the need to route assets through API Gateway.
- Free tier: 1 TB transfer + 10M requests/month — this site likely fits inside it.

Estimated saving: **$25–$40 / month**

② Remove API Gateway; serve via the ALB (or CloudFront) directly MEDIUM

You already pay for an Application Load Balancer ($1.70). API Gateway in front of a website is redundant and charged per-request. Point the domain at CloudFront → ALB → EC2 and delete the API Gateway stage.

Estimated saving: **~$21 / month** (most of the API Gateway line).

③ Remove the NAT Gateway (the VPC charge) MEDIUM

A single public-facing web/CMS instance does not need a NAT Gateway. Place the instance in a public subnet with a security group, or use a VPC endpoint for S3. If outbound internet is genuinely required, a small NAT instance is far cheaper than a managed NAT Gateway.

Estimated saving: **~$10 / month**

④ Compress & right-size all media (one-time cleanup) EASY

- **Videos:** re-encode reels to H.264/H.265 720p–1080p at a sane bitrate, or move them to YouTube/Vimeo embeds. 16 MB → ~2–3 MB each. (157 MB → ~25 MB.)
- **Images:** convert to WebP/AVIF and resize to display dimensions. The 193 MB portfolio + 98 MB product folders can shrink by 70–90%.
- Lazy-load below-the-fold media so it isn't fetched until needed.

This compounds with ①: less data to store, cache and transfer. Estimated extra saving: **$5–$15 / month** and faster page loads.

⑤ Right-size / schedule the EC2 instance MEDIUM

- Confirm the instance type. A static-ish Next.js + Strapi site usually runs comfortably on a `t3.small` / `t4g.small` (ARM/Graviton is ~20% cheaper).
- Buy a **1-year Savings Plan / Reserved Instance** once the size is settled → ~30–40% off the compute.

- Clean the 4 GB `.next/cache` on deploy and keep EBS volumes small.

Estimated saving: **$10–$20 / month**

⑥ Strongly consider hosting the frontend on Vercel / Amplify OPTIONAL

The app already includes `@vercel/analytics` and `@vercel/speed-insights`. Deploying the Next.js frontend on Vercel's free/hobby or low-cost tier removes EC2, ELB, API Gateway and NAT entirely for the frontend — leaving only a small instance (or managed host) for Strapi. This is the single largest structural saving if acceptable.

Potential saving: **$40–$70 / month** (replaces most of the AWS stack).

5. Projected bill after changes

SCENARIO	WHAT YOU DO	EST. MONTHLY	SAVING
Quick wins	CDN (①) + drop API Gateway (②) + drop NAT (③) + compress media (④)	$45–$60	**~$75–$90**
Optimised AWS	Quick wins + right-size EC2 & Savings Plan (⑤)	$30–$40	**~$95–$105**
Re-platform	Frontend on Vercel/Amplify (⑥) + tiny Strapi host	$15–$30	**~$105–$120**

Tax (~18%) scales down automatically with the base cost in every scenario.

6. Suggested order of execution

1. **Week 1 (zero risk):** Compress videos & images (④); add CloudFront in front of S3 (①). No app-logic change.
2. **Week 2:** Route the domain through CloudFront → ALB and delete the API Gateway stage (②).
3. **Week 3:** Move the instance to a public subnet / S3 VPC endpoint and remove the NAT Gateway (③).
4. **Week 4:** Right-size EC2 and purchase a Savings Plan (⑤). Evaluate Vercel for the frontend (⑥).

Before deleting anything: enable **AWS Cost Explorer** and turn on the **free Cost Anomaly Detection** alert so a future spike (like May→June) is caught within a day. Tag all resources with a project tag so each line item is traceable.

KMF AWS Cost Reduction Report — generated 24 June 2026 from static analysis of the project source (Next.js 14 + Strapi). Dollar estimates are derivid from the June 2026 AWS billing breakdown and standard AWS pricing; actual savings depend on traffic and the exact instance/NAT/API-Gateway configuration, which should be confirmed in the AWS console. No application code was modified in producing this report.